

Bent functions using Maiorana-McFarland secondary construction

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Abstract Bent functions are balanced by restricting their domains to vectors with either even or odd Hamming weights, which ensures an equal number of pre-images for both, 0 and 1. Using the previous fact, we can construct bent functions on two affine spaces. Additionally, we employ the Maiorana-McFarland secondary construction to obtain bent functions. Furthermore, the functions produced are balanced when we restrict their domain to vectors with even Hamming weights. We also conduct specific comparisons of the Walsh-Hadamard transform in relation to both the original bent functions and the newly generated ones. With all the necessary tools, we present an algorithm that leverages the Maiorana-McFarland secondary construction approach repeatedly to generate bent functions of any dimension.

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1 Introduction

Boolean functions $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ are essential in the fields of cryptography and coding theory. They possess important properties, including non-linearity, balancedness, low auto-correlation, and high algebraic immunity. The search

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space for these functions is vast, 2^{2^n} , making it challenging to identify suitable candidates. Different methods have been developed to find these functions, including random search, algebraic techniques, and heuristic methods, as discussed in sources [14] and [12].

We are interested in the non-linearity of a boolean function, defined as the distance between this and the set of affine functions. The boolean functions with maximum non-linearity are called bent functions, and Rothaus introduced this name in 1976 [10]. These functions have been classified and constructed in many ways, such as the Maiorana-McFarland class [6] and Rothaus [10]. In some studies, the balancedness of a boolean function is restricted to specific subsets of its domain based on Hamming weight [4]. We analyse the balancedness of a bent function when its domain is restricted to vectors with either even or odd Hamming weight.

In this work, we use a particular case of the Maiorana-McFarland secondary construction [5], [6]: given a function $\phi : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$ such that $\phi^{-1}(a)$ is an affine space of dimension $s - 1$ and a function $g_e : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$, such that $g_e|_{\phi^{-1}(a)}$ is a bent function, then $f : \mathbb{F}_2^{s+1} \rightarrow \mathbb{F}_2$ is a bent function, where $\bar{x} \mapsto \bar{x} \cdot \phi(\bar{y}) \oplus g_e(\bar{y})$.

In [1], it is shown that a bent function defined over the domain $\mathbb{F}_2^n$ is balanced when restricted to one of the two linear hyperplanes that form a partition of the domain. We provide an alternative proof by focusing specifically on affine spaces containing vectors with even and odd Hamming weights. We then identify bent functions g_{e_i} for $i = 0, 1$, using these spaces as their domains, based on the relation $f(\bar{x}, \bar{y}) = g(\bar{x})$, where g is a bent function. To demonstrate this result, we can apply Theorem V.2 from [1], or use a more straightforward approach by examining the balancedness of elements with even and odd Hamming weights.

Affine spaces can be utilised alongside boolean functions to analyse the decomposition and the called normal functions [2], [11]. Additionally, affine spaces can also be seen in [15], where the relation $f(\bar{x}, \bar{y}) = g(\bar{x})$ is employed, along with other conditions, to derive plateaued functions. These functions are instrumental in constructing bent functions via their Construction K. [3] focuses on functions $f : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$, where p is a prime number. In this context, the relation $f(\bar{x}, \bar{y}) = g(\bar{x})$ is again applied, demonstrating that $f(\bar{x}, \bar{y})$ is a partially bent function if g is a bent function.

We proceed as follows: In Section 2, we review the preliminaries. Then, in Section 3, we demonstrate that a bent function is balanced when we restrict its domain to either the set of vectors with even Hamming weight or the set with odd Hamming weight. This process illustrates the distribution of the number of pre-images for both 0 and 1 across the entire domain of the bent function. In Section 4, we recall and provide a demonstration regarding bent functions over affine spaces: boolean functions g_{e_0} and g_{e_1} over affine spaces of dimension $s - 1$ are defined and demonstrated to be bent functions. Subsequently, we define the necessary elements $\phi : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$, $g_e : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$, a previous bent function $g : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$, and the affine spaces to construct the new Maiorana bent function $f : \mathbb{F}_2^{1+s} \rightarrow \mathbb{F}_2$. Furthermore, f is uniquely

balanced when we restrict its domain to the set of vectors with even Hamming weight. Additionally, we analyse the balancedness of g_{e_0} , g_{e_1} , and $f \oplus l_{\bar{a}}$, where $l_{\bar{a}}$ is a linear function defined as $l_{\bar{a}}(\bar{x}) = \bar{a} \cdot \bar{x}$, $\bar{a}, \bar{x} \in \mathbb{F}_2^{1+s}$. Finally, we provide an algorithm to construct a new bent function of any dimension by repeatedly applying Theorem 10.

2 Background

Definitions and results about boolean functions, particularly bent functions, are recalled in this section. These can be found, for example, in [8], [9], [7].

Definition 1 A function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is called a **boolean function**. $\mathcal{B}_n$ is the set of all boolean functions with domain $\mathbb{F}_2^n$.

All boolean functions $f \in \mathcal{B}_n$ have an **algebraic normal form** (ANF):

$$f(\bar{x}) = \bigoplus_{\bar{u} \in \mathbb{F}_2^n} a_{\bar{u}} \bar{x}^{\bar{u}},$$

$$a_{\bar{u}} \in \mathbb{F}_2, \bar{x}^{\bar{u}} = x_1^{u_1} \cdots x_n^{u_n}, \quad \bar{x} = (x_1, \dots, x_n), \bar{u} = (u_1, \dots, u_n).$$

Example 1 The boolean function $f \in \mathcal{B}_3$, $f(x_1, x_2, x_3) = 1 \oplus x_1x_2 \oplus x_1x_2x_3$ is in its ANF.

Theorem 1 Let $f \in \mathcal{B}_n$. Then,

$$f(\bar{x}) = \bigoplus_{\bar{u} \in \mathbb{F}_2^n} a_{\bar{u}} \bar{x}^{\bar{u}},$$

$$a_{\bar{u}} = \bigoplus_{\bar{x} \leq \bar{u}} f(\bar{x}), \quad \bar{x} \leq \bar{u} \Leftrightarrow x_i \leq u_i, \quad \bar{x} = (x_1, \dots, x_n), \bar{u} = (u_1, \dots, u_n).$$

Definition 2 The set of all **affine** boolean functions with domain $\mathbb{F}_2^n$, denoted by $\mathcal{A}_n$, is defined as

$$\mathcal{A}_n := \{l_{\bar{a}, a_0} \mid \bar{a} \in \mathbb{F}_2^n, a_0 \in \mathbb{F}_2\},$$

where $l_{\bar{a}, a_0}(x) = \bar{a} \cdot \bar{x} \oplus a_0$ and $\cdot$ is the dot product.

We will also use the notation $l_{\bar{a}}$ when the affine function $l_{\bar{a}, a_0}$ has a_0 equal to zero. In this case, $l_{\bar{a}}$ is called a linear function. Note that the number of affine functions is 2^{n+1} and the number of linear functions is 2^n .

Definition 3 The **non-linearity** of a boolean function $f \in \mathcal{B}_n$ is defined as the Hamming distance between f and the set of affine functions:

$$Nl(f) := \min_{g \in \mathcal{A}_n} d_H(f, g).$$

Boolean functions that achieve the maximum non-linearity are called **bent** functions.

We define the following function to characterize the non-linearity:

Definition 4 The **Walsh-Hadamard Transform** of a boolean function $f \in \mathcal{B}_n$ is defined as

$$\widehat{\mathcal{W}}_f(\bar{a}) = \sum_{\bar{x} \in \mathbb{F}_2^n} (-1)^{f(\bar{x}) \oplus \bar{a} \cdot \bar{x}}, \quad \bar{a} \in \mathbb{F}_2^n.$$

Theorem 2 *The non-linearity of the boolean function $f \in \mathcal{B}_n$ is characterized as*

$$Nl(f) = 2^{n-1} - \frac{1}{2} \max_{\bar{a} \in \mathbb{F}_2^n} |\widehat{\mathcal{W}}_f(\bar{a})|.$$

Theorem 3 *The function $f \in \mathcal{B}_n$ is consider bent if and only if $\widehat{\mathcal{W}}_f(\bar{a}) = \pm 2^{n/2}$ for every $\bar{a} \in \mathbb{F}_2^n$.*

The bent functions have non-linearity $2^{n-1} - 2^{n/2-1}$.

3 The balancedness of bent functions in two affine subspaces of their domain

The following result can be found in [1], where it is proven for any affine subspace of $\mathbb{F}_2^n$ with dimension $n-1$ and its affine complement. The authors employ the derivative of a Boolean function to demonstrate this. In this work, we provide an alternative direct proof specifically for the case of the affine space containing elements with even Hamming weight and its complement, which consists of elements with odd Hamming weight.

Theorem 4 *Let $\mathcal{A} := \{\bar{x} \in \mathbb{F}_2^n | w_H(\bar{x}) \text{ is even}\}$ and $\mathcal{B} := \{\bar{x} \in \mathbb{F}_2^n | w_H(\bar{x}) \text{ is odd}\}$. Every bent function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, where $n \geq 2$ and n is even, satisfies that either $f|_{\mathcal{A}}$ or $f|_{\mathcal{B}}$ is balanced.*

Proof Let f be a bent function. We know that

$$w_H(f \oplus l_{\bar{a}}) = 2^{n-1} - 2^{\frac{n-2}{2}} \text{ or } w_H(f \oplus l_{\bar{a}}) = 2^{n-1} + 2^{\frac{n-2}{2}},$$

where $\bar{a} \in \mathbb{F}_2^n$.

Observe that, $l_{\bar{0}}(\bar{x}) = 0$, $l_{\bar{1}}(\bar{x}) = x_1 \oplus \dots \oplus x_n$, $l_{\bar{1}}(\mathcal{A}) = \{0\}$, and $l_{\bar{1}}(\mathcal{B}) = \{1\}$. Here, $\bar{x}, \bar{0}, \bar{1} \in \mathbb{F}_2^n$, where $\bar{x} = (x_1, \dots, x_n)$, $\bar{0} = (0, \dots, 0)$, and $\bar{1} = (1, \dots, 1)$.

First, we assume that $n \geq 6$.

Case 1. If $w_H(f|_{\mathcal{A}}) = c$, $0 \leq c \leq 2^{\frac{n-2}{2}}$.

Case 1a. $w_H(f|_{\mathcal{B}}) = 2^{n-1} - (2^{\frac{n-2}{2}} + c)$. Then, $w_H((f \oplus l_{\bar{1}})|_{\mathcal{A}}) = c$ and $w_H((f \oplus l_{\bar{1}})|_{\mathcal{B}}) = 2^{\frac{n-2}{2}} + c$.

Case 1a1 $w_H(f \oplus l_{\bar{1}}) = c + (2^{\frac{n-2}{2}} + c) = 2^{n-1} - 2^{\frac{n-2}{2}}$. Then, $c = 2^{n-2} - 2^{\frac{n-2}{2}}$. Since $n \geq 6$, it follows that $2^{\frac{n-2}{2}} < c$. This leads a contradiction.

Case 1a2 $w_H(f \oplus l_{\bar{1}}) = c + (2^{\frac{n-2}{2}} + c) = 2^{n-1} + 2^{\frac{n-2}{2}}$. Then, $c = 2^{n-2}$. This implies that, $2^{\frac{n-2}{2}} < c$. As a result, we encounter a contradiction.

Case 1b $w_H(f|_{\mathcal{B}}) = 2^{n-1} + (2^{\frac{n-2}{2}} - c)$. Thus, we find that $c = 2^{\frac{n-2}{2}}$, which results in $w_H((f \oplus l_{\bar{1}})|_{\mathcal{A}}) = 2^{\frac{n-2}{2}}$ and $w_H((f \oplus l_{\bar{1}})|_{\mathcal{B}}) = 0$. Therefore, $f \oplus l_{\bar{1}}$ cannot be a bent function when $n \geq 6$, leading us to a contradiction.

Case 2. If $w_H(f|_{\mathcal{A}}) = c + 2^{\frac{n-2}{2}}$, $0 < c \leq 2^{n-2} - 2^{\frac{n-2}{2}}$.

Case 2a. $w_H(f|_{\mathcal{B}}) = 2^{n-2} + (2^{n-2} - 2 \cdot 2^{\frac{n-2}{2}} - c)$. Then, $w_H((f \oplus l_{\bar{1}})|_{\mathcal{A}}) = c + 2^{\frac{n-2}{2}}$ and $w_H((f \oplus l_{\bar{1}})|_{\mathcal{B}}) = 2 \cdot 2^{\frac{n-2}{2}} + c$.

Case 2a1 $w_H(f \oplus l_{\bar{1}}) = c + 2^{\frac{n-2}{2}} + 2 \cdot 2^{\frac{n-2}{2}} + c = 2^{n-1} - 2^{\frac{n-2}{2}}$. Thus, we find that $c = 2^{n-2} - 2^{\frac{n-2}{2}}$, which results in $w_H(f|_{\mathcal{B}}) = 2^{n-2}$. Therefore, $f|_{\mathcal{B}}$ is balanced, and there is no contradiction.

Case 2a2 $w_H(f \oplus l_{\bar{1}}) = c + 2^{\frac{n-2}{2}} + 2 \cdot 2^{\frac{n-2}{2}} + c = 2^{n-1} + 2^{\frac{n-2}{2}}$. Thus, we find that $c = 2^{n-2} - 2^{\frac{n-2}{2}}$, which results in $w_H(f|_{\mathcal{A}}) = 2^{n-2}$. Therefore, $f|_{\mathcal{A}}$ is balanced, and there is no contradiction.

Case 2b $w_H(f|_{\mathcal{B}}) = 2^{n-2} + (2^{n-2} - c)$. Then, $w_H((f \oplus l_{\bar{1}})|_{\mathcal{A}}) = c + 2^{\frac{n-2}{2}}$ and $w_H((f \oplus l_{\bar{1}})|_{\mathcal{B}}) = c$.

Case 2b1 $w_H(f \oplus l_{\bar{1}}) = c + (c + 2^{\frac{n-2}{2}}) = 2^{n-1} - 2^{\frac{n-2}{2}}$. Thus, we find that $c = 2^{n-2} - 2^{\frac{n-2}{2}}$, which results in $w_H(f|_{\mathcal{A}}) = 2^{n-2}$. Therefore, $f|_{\mathcal{A}}$ is balanced, and there is no contradiction.

Case 2b2 $w_H(f \oplus l_{\bar{1}}) = c + (c + 2^{\frac{n-2}{2}}) = 2^{n-1} + 2^{\frac{n-2}{2}}$. Then, $c = 2^{n-2}$. This implies that, $0 < c \leq 2^{n-2} - 2^{\frac{n-2}{2}}$. As a result, we encounter a contradiction.

When $w_H(f|_{\mathcal{A}}) > 2^{n-2}$, the proof is similar to the previous cases. However, we consider the number of images 0 of $f|_{\mathcal{A}}$ instead of the number of images 1. We also use the fact that $w_H(l_{\bar{1}} \oplus f) = 2^{n-1} - 2^{\frac{n-2}{2}}$ or $w_H(l_{\bar{1}} \oplus f) = 2^{n-1} + 2^{\frac{n-2}{2}}$. This implies that the number of images 0 of f is either $2^{n-1} - 2^{\frac{n-2}{2}}$ or $2^{n-1} + 2^{\frac{n-2}{2}}$.

Among all possible cases, contradictions arise only when both $f|_{\mathcal{A}}$ and $f|_{\mathcal{B}}$ are unbalanced. Therefore, if f is a bent function, either $f|_{\mathcal{A}}$ or $f|_{\mathcal{B}}$ must be balanced.

Case $n = 2$ is direct, and case $n = 4$ only needs light observations. $\square$

Due to the previous result, the cardinality distribution of the preimages of a bent function is as follows:

Remark 1 Let $g : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be a bent function such that $g|_{\mathcal{A}}$ is balanced.

(a) If $\widehat{\mathcal{W}}_g(\bar{0}) = 2^{\frac{n}{2}}$, then

$$|(g)_{|\mathcal{A}}^{-1}(0)| = 2^{n-2}$$

$$|(g)_{|\mathcal{A}}^{-1}(1)| = 2^{n-2}$$

$$|(g)_{|\mathcal{B}}^{-1}(0)| = 2^{n-2} + 2^{\frac{n-2}{2}}.$$

$$|(g)_{|\mathcal{B}}^{-1}(1)| = 2^{n-2} - 2^{\frac{n-2}{2}}.$$

(b) If $\widehat{\mathcal{W}}_g(\bar{0}) = -2^{\frac{n}{2}}$, then

$$|(g)_{|\mathcal{A}}^{-1}(0)| = 2^{n-2}$$

$$|(g)_{|\mathcal{A}}^{-1}(1)| = 2^{n-2}$$

$$|(g)_{|\mathcal{B}}^{-1}(0)| = 2^{n-2} - 2^{\frac{n-2}{2}}.$$

$$|(g)_{|\mathcal{B}}^{-1}(1)| = 2^{n-2} + 2^{\frac{n-2}{2}}.$$

We obtain a similar observation if $g|_{\mathcal{B}}$ is balanced.

4 Construction of a particular family from the Maiorana-McFarland secondary construction

We broaden the definition of bent functions with domain $\mathbb{F}_2^n$ to include bent functions defined on an affine subspace, as suggested in Proposition 1 of [6].

We denote the set of all affine boolean functions with domain $\mathcal{C} \subseteq \mathbb{F}_2^n$ by $\mathcal{A}_n^{\mathcal{C}}$. Note that linear functions of $\mathcal{A}_n^{\mathcal{C}}$ can be expressed as $l_{\bar{a}, a_0|_{\mathcal{C}}}$, where $\bar{a}$ is any element of $\mathbb{F}_2^n$ and a_0 is any element of $\mathbb{F}_2$.

Definition 5 A function $f : \mathcal{C} \rightarrow \mathbb{F}_2$, $\mathcal{C} \subseteq \mathbb{F}_2^n$ an affine space, $m \leq n$, $\dim \mathcal{C} = m$, is bent if $Nl(f) = d_H(f, \mathcal{A}_n^{\mathcal{C}})$ achieve the maximum.

The following results are straightforward to obtain; the demonstrations are similar to traditional boolean function proofs.

Theorem 5 Let a function $f : \mathcal{C} \rightarrow \mathbb{F}_2$. Then

$$Nl(f) = 2^{m-1} - \frac{1}{2} \max_{\bar{a} \in \mathbb{F}_2^n} |\widehat{\mathcal{W}}_f(\bar{a})|,$$

$$\text{where } \widehat{\mathcal{W}}_f(\bar{a}) := \sum_{\bar{x} \in \mathcal{C}} (-1)^{f(\bar{x}) \oplus \bar{a} \cdot \bar{x}}.$$

Proof We can see that, for all $\bar{a} \in \mathbb{F}_2^n$,

$$\widehat{\mathcal{W}}_f(\bar{a}) = 2^m - 2d_H(f, \bar{a} \cdot \bar{x}) \text{ and } -\widehat{\mathcal{W}}_f(\bar{a}) = 2^m - 2d_H(f, \bar{a} \cdot \bar{x} \oplus 1).$$

Resolving d_H on the left side, we obtain the result in both cases.

□

Theorem 6 (Parseval's equation) *Let $f : \mathcal{C} \rightarrow \mathbb{F}_2$. Then,*

$$\sum_{\bar{a} \in \mathbb{F}_2^n} \widehat{\mathcal{W}}_f^2(\bar{a}) = 2^{m+n}.$$

Proof Resolving,

$$\sum_{\bar{a} \in \mathbb{F}_2^n} \widehat{\mathcal{W}}_f(\bar{a}) \widehat{\mathcal{W}}_f(\bar{a}) = \sum_{\bar{x}, \bar{y} \in \mathcal{C}} (-1)^{f(\bar{x}) \oplus f(\bar{y})} \sum_{\bar{a} \in \mathbb{F}_2^n} (-1)^{\bar{a} \cdot (\bar{x} \oplus \bar{y})} = 2^{m+n}.$$

□

Theorem 7 *The function $f : \mathcal{C} \subset \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, $\dim \mathcal{C} = m$, is a bent if and only if $\widehat{\mathcal{W}}_f(\bar{a}) = \pm 2^{m/2}$ for all $\bar{a} \in \mathbb{F}_2^n$.*

Proof If $|\widehat{\mathcal{W}}_f(\bar{b})| < 2^{m/2}$, then by Parseval's equation, exists $\bar{b}' \in \mathbb{F}_2^n$ such that $|\widehat{\mathcal{W}}_f(\bar{b}')| > 2^{m/2}$. Hence, according to Theorem 5, the non-linearity of f is the greatest when $\widehat{\mathcal{W}}_f(\bar{a}) = \pm 2^{m/2}$ for all $\bar{a} \in \mathbb{F}_2^n$.

□

The following theorem corresponds to a class of bent functions known as **Maiorana-McFarland secondary construction**.

Theorem 8 [6] *Let the function $\phi(\bar{y}) : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^r$ be defined such that for all $\bar{a} \in \mathbb{F}_2^r$, the preimage $\phi^{-1}(\bar{a})$ forms an affine space of dimension $s-r$. Additionally, let $g_e(\bar{y}) : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$ be a function for which the restriction $g_e|_{\phi^{-1}(\bar{a})}$ is a bent function. We then define the function $f : \mathbb{F}_2^{r+s} \rightarrow \mathbb{F}_2$ by $f(\bar{x}, \bar{y}) = \bar{x} \cdot \phi(\bar{y}) \oplus g_e(\bar{y})$, where $\bar{x} \in \mathbb{F}_2^r$. Under these conditions, the function f is bent.*

Let $\mathcal{C}_0 := \{\bar{x} \in \mathbb{F}_2^s \mid w_H(\bar{x}) \text{ even}\}$ and $\mathcal{C}_1 := \{\bar{x} \in \mathbb{F}_2^s \mid w_H(\bar{x}) \text{ odd}\}$. In this work, we consider $\mathbb{F}_2^s$ as an array where each element represents a row. We will order the elements in the following way:

$$\mathbb{F}_2^s = \begin{array}{cc} & \begin{array}{c} \mathcal{C}'_0 \quad \bar{0} \\ \mathcal{C}'_1 \quad \bar{1} \end{array} \\ \begin{array}{c} \mathcal{C}_0 \\ \mathcal{C}_1 \end{array} & \begin{array}{c} \text{---} \\ \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \mathcal{C}'_0 \quad \bar{1} \\ \mathcal{C}'_1 \quad \bar{0} \end{array}, \end{array}$$

where $\bar{0} = \begin{array}{c} 0 \\ \vdots \\ 0 \end{array}$ and $\bar{1} = \begin{array}{c} 1 \\ \vdots \\ 1 \end{array}$ are $2^{s-2} \times 1$ arrays, and

$$\mathcal{C}'_0 := \{\bar{x} \in \mathbb{F}_2^{s-1} \mid w_H(\bar{x}) \text{ even}\}, \quad \mathcal{C}'_1 := \{\bar{x} \in \mathbb{F}_2^{s-1} \mid w_H(\bar{x}) \text{ odd}\}.$$

Notice that $\mathcal{C}_0$ is a linear code with dimension $s-1$, and that $\mathcal{C}_1$ is an affine space such that for any $\bar{b} \in \mathbb{F}_2^s$ with an odd Hamming weight, we have $\bar{b} \oplus \mathcal{C}_0 = \mathcal{C}_1$.

From now on, we will consider a bent function $g : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$, according to Theorem 8 (for the particular case $r = 1$). We define $\phi : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$ and $g_e : \mathbb{F}_2^s \rightarrow \mathbb{F}_2$ as follows:

$$\phi^{-1}(0) = \mathcal{C}_0 \text{ and } \phi^{-1}(1) = \mathcal{C}_1. \text{ That is, } \phi = l_{\bar{1}}, \bar{1} = (1, \dots, 1),$$

$$g_{e|_{\mathcal{C}_0}} := g_{e_0} \text{ and } g_{e|_{\mathcal{C}_1}} := g_{e_1} \text{ so that,}$$

$$g_{e_0} : \mathcal{C}_0 \rightarrow \mathbb{F}_2, g_{e_0}(\bar{x}, x_s) := g(\bar{x}), \bar{x} \in \mathbb{F}_2^{s-1}, x_s \in \mathbb{F}_2,$$

$$g_{e_1} : \mathcal{C}_1 \rightarrow \mathbb{F}_2, g_{e_1}(\bar{x}, x_s) := g(\bar{x}), \bar{x} \in \mathbb{F}_2^{s-1}, x_s \in \mathbb{F}_2.$$

Remark 2 In $\mathcal{C}_0$, if $\bar{x} \in \mathbb{F}_2^{s-1}$ has an even Hamming weight, then x_s is 0. If $\bar{x} \in \mathbb{F}_2^{s-1}$ has an odd Hamming weight, then $x_s \in \mathbb{F}_2$ is 1. The same applies in $\mathcal{C}_1$.

The following theorem can be proven using from [1, Theorem V.2], as g_e meets the condition $\mathcal{H}_\alpha$ with $\alpha = 1$ referenced in [1]. In that theorem, the authors utilize the derivative of a Boolean function to support their claims.

We provide a direct proof demonstrating that the restricted balancedness of a bent function is essential.

Theorem 9 *Let $g : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function. Then, $g_{e_0} : \mathcal{C}_0 \rightarrow \mathbb{F}_2$ is a bent function and $g_{e_1} : \mathcal{C}_1 \rightarrow \mathbb{F}_2$ is a bent function.*

Proof Let $\bar{a} = (a, a_s) \in \mathbb{F}_2^s$, $\bar{x} = (x, x_s) \in \mathcal{C}_0$, $l_a(x) = a_1x_1 + \dots + a_{s-1}x_{s-1}$, where $a = (a_1, \dots, a_{s-1})$ and $x = (x_1, \dots, x_{s-1})$.

$$\widehat{\mathcal{W}}_{g_{e_0}}(\bar{a}) = \sum_{\bar{x} \in \mathcal{C}_0} (-1)^{g_{e_0}(\bar{x}) \oplus \bar{x} \cdot \bar{a}} = \sum_{\bar{x} \in \mathcal{C}_0} (-1)^{g_{e_0}(x, x_s) \oplus x \cdot a \oplus x_s a_s}.$$

If $a_s = 0$,

$$\widehat{\mathcal{W}}_{g_{e_0}}(\bar{a}) = \sum_{x \in \mathbb{F}_2^{s-1}} (-1)^{g(x) \oplus x \cdot a} = \widehat{\mathcal{W}}_g(a).$$

If $a_s = 1$,

$$\begin{aligned} \widehat{\mathcal{W}}_{g_{e_0}}(\bar{a}) &= \sum_{\bar{x} \in \mathcal{C}_0} (-1)^{g_{e_0}(x, 0) \oplus x \cdot a} + \sum_{\bar{x} \in \mathcal{C}_0} (-1)^{g_{e_0}(x, 1) \oplus x \cdot a \oplus 1} \\ &= \sum_{x \in \mathcal{C}'_0} (-1)^{g(x) \oplus x \cdot a} + (-1) \sum_{x \in \mathcal{C}'_1} (-1)^{g(x) \oplus x \cdot a}. \end{aligned}$$

The last equality, by Remark 2.

If $(g \oplus l_a)|_{\mathcal{C}'_0}$ is balanced, $\widehat{\mathcal{W}}_{g_{e_0}}(\bar{a}) = -\widehat{\mathcal{W}}_g(a)$.

If $(g \oplus l_a)|_{\mathcal{C}'_1}$ is balanced, $\widehat{\mathcal{W}}_{g_{e_0}}(\bar{a}) = \widehat{\mathcal{W}}_g(a)$.

In both cases, since g is a bent function, g_{e_0} is a bent function.

Proceeding similarly, $g_{e_1} : \mathcal{C}_1 \rightarrow \mathbb{F}_2$ is a bent function.

□

Corollary 1 Let $g : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function. For all $a \in \mathbb{F}_2^{s-1}$:

1. $\widehat{\mathcal{W}}_g(a) = \widehat{\mathcal{W}}_{g_{e_0}}(a, 0) = \widehat{\mathcal{W}}_{g_{e_1}}(a, 0)$.
2. If $(g \oplus l_a)_{|C'_0}$ is balanced,
 $\widehat{\mathcal{W}}_g(a) = 2^{\frac{s-1}{2}} \Leftrightarrow \widehat{\mathcal{W}}_{g_{e_0}}(a, 1) = -2^{\frac{s-1}{2}}$ and $\widehat{\mathcal{W}}_{g_{e_1}}(a, 1) = 2^{\frac{s-1}{2}}$.
3. If $(g \oplus l_a)_{|C'_1}$ is balanced,
 $\widehat{\mathcal{W}}_g(a) = -2^{\frac{s-1}{2}} \Leftrightarrow \widehat{\mathcal{W}}_{g_{e_0}}(a, 1) = 2^{\frac{s-1}{2}}$ and $\widehat{\mathcal{W}}_{g_{e_1}}(a, 1) = -2^{\frac{s-1}{2}}$.

□

Functions g_{e_0} and g_{e_1} , like function g , are balanced when their domain is restricted. The balancedness is maintained even when we add a linear function.

Corollary 2 Let $g : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function and $\bar{a} = (a, a_s)$, $a \in \mathbb{F}_2^{s-1}$, $a_s \in \mathbb{F}_2$. Then, for all $\bar{a} \in \mathbb{F}_2^s$, $g_{e_i} \oplus l_{\bar{a}} : C_i \rightarrow \mathbb{F}_2$ is a bent function, $i = 0, 1$. If $(g \oplus l_a)_{|C'_0}$ is balanced, $g_{e_0} \oplus l_{(a, a_s)}$ is balanced restricted to $C'_0|0$ and $g_{e_1} \oplus l_{(a, a_s)}$ is balanced restricted to $C'_0|1$. If $(g \oplus l_a)_{|C'_1}$ is balanced, then $g_{e_0} \oplus l_{(a, a_s)}$ is balanced restricted to $C'_1|1$ and $g_{e_1} \oplus l_{(a, a_s)}$ is balanced restricted to $C'_1|0$.

Proof Let us see first that the desired functions are bent:

Given that $\widehat{\mathcal{W}}_{g_{e_i}}(\bar{b}) = \pm 2^{\frac{s-1}{2}}$, $i = 0, 1$, for all $\bar{b} \in \mathbb{F}_2^s$, and the linearity of the linear functions, then $g_{e_i} \oplus l_{\bar{a}}$, $i = 0, 1$, is a bent function for all $\bar{a} \in \mathbb{F}_2^s$.

Now, we prove the balancedness:

Since $g \oplus l_a$ is a bent function, without loss of generality, suppose that $g \oplus l_a : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$, $a \in \mathbb{F}_2^{s-1}$, is balanced restricted to C'_0 . Hence, given that the value $a_s \in \mathbb{F}_2$ is constant and the images of g , g_{e_0} , and g_{e_1} are equal, then $g_{e_0} \oplus l_{(a, a_s)}$ is balanced restricted to $C'_0|0$ and $g_{e_1} \oplus l_{(a, a_s)}$ is balanced restricted to $C'_0|1$.

It is solved in a similar way, if the balancedness restricted to C'_1 is considered.

□

Now, we are ready to use the Maiorana-McFarland secondary construction, in the particular case where $r = 1$. The following theorem considers ϕ and g_e as described on page 7.

Theorem 10 Let $g : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function. Then, the function

$$f : \mathbb{F}_2^{1+s} \rightarrow \mathbb{F}_2, (x_0, \bar{x}) \mapsto x_0\phi(\bar{x}) \oplus g_e(\bar{x}), x_0 \in \mathbb{F}_2, \bar{x} \in \mathbb{F}_2^s,$$

is a bent function and $f_{| \{(x_0, \bar{x}) | w_H((x_0, \bar{x})) \text{ even} \}}$ is balanced.

Proof Since ϕ and g_e satisfy the conditions of Theorem 8, then f is a bent function.

Let us examine the balancedness of f . Let $\bar{x} = (x, x_s)$, where $x \in \mathbb{F}_2^{s-1}$, $x_s \in \mathbb{F}_2$. We know that

$$f(x_0, x, x_s) = x_0\phi(x, x_s) \oplus g_e(x, x_s).$$

We have four cases. Without loss of generality, consider the case when $g|_{\mathcal{C}'_0}$ is balanced and $\widehat{\mathcal{W}}_g(\bar{0}) = 2^{\frac{s-1}{2}}$.

The elements with even Hamming weight in $\mathbb{F}_2^{s+1}$ are in the following cases:

Case 1. Let $x_0 = 0$, $x_s = 0$, and $x \in \mathcal{C}'_0$. Then,
 $w_H(f|_{(0, \mathcal{C}'_0, 0)}) = w_H(g_{e_0}|_{(\mathcal{C}'_0, 0)}) = 2^{s-3}$.

Case 2. Let $x_0 = 0$, $x_s = 1$, and $x \in \mathcal{C}'_1$. Then,
 $w_H(f|_{(0, \mathcal{C}'_1, 1)}) = w_H(g_{e_0}|_{(\mathcal{C}'_1, 1)}) = 2^{s-3} - 2^{\frac{s-3}{2}}$.

Case 3. Let $x_0 = 1$, $x_s = 0$, and $x \in \mathcal{C}'_1$. Then,
 $w_H(f|_{(1, \mathcal{C}'_1, 0)}) = w_H((1 \oplus g_{e_1})|_{(\mathcal{C}'_1, 0)}) = 2^{s-3} + 2^{\frac{s-3}{2}}$.

Case 4. Let $x_0 = 1$, $x_s = 1$, and $x \in \mathcal{C}'_0$. Then,
 $w_H(f|_{(1, \mathcal{C}'_0, 1)}) = w_H((1 \oplus g_{e_1})|_{(\mathcal{C}'_0, 1)}) = 2^{s-3}$.

Hence, adding the four cases,

$$|f^{-1}_{|\{(x_0, \bar{x}) \in \mathbb{F}_2^{s+1} | w_H(x_0, \bar{x}) \text{ even}\}}(1)| = 2^{s-1}.$$

Therefore, $f|_{\{(x_0, \bar{x}) | w_H((x_0, \bar{x})) \text{ even}\}}$ is balanced.

In a similar manner, we achieved the same outcome in the other cases as well. □

Given a balanced bent function when we restrict to vectors of even Hamming weight (using Maiorana-McFarland secondary construction), we want to know about the balancedness if we add a linear function. We use principally the functions considered in Corollary 2, applying Remark 1.

Theorem 11 *Let $g : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function and*

$$f : \mathbb{F}_2^{s+1} \rightarrow \mathbb{F}_2, (x_0, \bar{x}) \mapsto x_0 \phi(\bar{x}) \oplus g_e(\bar{x}), \quad x_0 \in \mathbb{F}_2, \quad \bar{x} \in \mathbb{F}_2^s,$$

be a Maiorana-McFarland (secondary construction) bent function. Then, the function $f \oplus l_{(a_0, \bar{a})}$, where $l_{(a_0, \bar{a})}(x_0, \bar{x}) = a_0 x_0 \oplus \bar{a} \cdot \bar{x}$, $a_0 \in \mathbb{F}_2$, $\bar{a} = (a_1, \dots, a_s) \in \mathbb{F}_2^s$, satisfy the following:

If $(a_0 = 0 \wedge a_s = 0) \vee (a_0 = 1 \wedge a_s = 1)$, then

$$(f \oplus l_{(a_0, \bar{a})})|_{\{(x_0, \bar{x}) | w_H(x_0, \bar{x}) \text{ even}\}} \text{ is balanced.}$$

If $(a_0 = 0 \wedge a_s = 1) \vee (a_0 = 1 \wedge a_s = 0)$, then

$$(f \oplus l_{(a_0, \bar{a})})|_{\{(x_0, \bar{x}) | w_H(x_0, \bar{x}) \text{ odd}\}} \text{ is balanced.}$$

Proof Let $\bar{a} = (a, a_s)$ and $\bar{x} = (x, x_s)$, $a, x \in \mathbb{F}_2^{s-1}$, $a_s, x_s \in \mathbb{F}_2$. We have four possibilities for the pair (a_0, a_s) . For each option, we will demonstrate the balancedness of the function $g_e \oplus l_{(a, a_s)}$ across the following sets:

$$\mathcal{C}'_0|\bar{0}, \mathcal{C}'_1|\bar{1}, \mathcal{C}'_0|\bar{1}, \text{ and } \mathcal{C}'_1|\bar{0}.$$

Additionally, we consider the corresponding values of $\widehat{\mathcal{W}}_{g_e \oplus l_{(a, a_s)}}(\bar{0})$, which can equal either $2^{\frac{s-1}{2}}$ or $-2^{\frac{s-1}{2}}$ in the respective complements of the affine space considered, specifically in:

$$\mathcal{C}'_1|\bar{1}, \mathcal{C}'_0|\bar{0}, \mathcal{C}'_1|\bar{0}, \text{ and } \mathcal{C}'_0|\bar{1}$$

respectively.

We know that,

$$(f \oplus l_{(a_0, a, a_s)})(x_0, x, x_s) = x_0 \phi(x, x_s) \oplus g_e(x, x_s) \oplus l_{(a, a_s)}(x, x_s) \oplus a_0 x_0.$$

Without loss of generality, let us see the case $a_0 = 0$ and $a_s = 1$ when $g_e \oplus l_{(a, 1)}$ is balanced in $\mathcal{C}'_0|\bar{1}$ and $\widehat{\mathcal{W}}_{g_e \oplus l_{(a, 1)}}(\bar{0}) = -2^{\frac{s-1}{2}}$ in $\mathcal{C}'_1|\bar{0}$.

The elements with odd Hamming weight in $\mathbb{F}_2^{1+s}$ are in the following cases:

Case 1. Let $x_0 = 0$, $x_s = 0$, and $w_H(x)$ odd. Then,

$$w_H((f \oplus l_{(0, a, 1)})|_{(0, \mathcal{C}'_1, 0)}) = w_H((g_{e_1} \oplus l_{(a, 1)})|_{(\mathcal{C}'_1, 0)}) = 2^{s-3} + 2^{\frac{s-3}{2}}.$$

Since, $g_e \oplus l_{(a, 1)}$ is not balanced and $\widehat{\mathcal{W}}_{g_e \oplus l_{(a, 1)}}(\bar{0}) = -2^{\frac{s-1}{2}}$ in $\mathcal{C}'_1|\bar{0}$.

Case 2. Let $x_0 = 0$, $x_s = 1$, and $w_H(x)$ even. Then,

$$w_H((f \oplus l_{(0, a, 1)})|_{(0, \mathcal{C}'_0, 1)}) = w_H((g_{e_1} \oplus l_{(a, 1)})|_{(\mathcal{C}'_0, 1)}) = 2^{s-3}.$$

Since, $g_e \oplus l_{(a, 1)}$ is balanced in $\mathcal{C}'_0|\bar{1}$.

Case 3. Let $x_0 = 1$, $x_s = 0$, and $w_H(x)$ even. Then,

$$w_H((f \oplus l_{(0, a, 1)})|_{(1, \mathcal{C}'_0, 0)}) = w_H((g_{e_0} \oplus l_{(a, 1)})|_{(\mathcal{C}'_0, 0)}) = 2^{s-3}.$$

Since, $g_e \oplus l_{(a, 1)}$ is balanced in $\mathcal{C}'_0|\bar{0}$.

Case 4. Let $x_0 = 1$, $x_s = 1$, and $w_H(x)$ odd. Then,

$$w_H((f \oplus l_{(0, a, 1)})|_{(1, \mathcal{C}'_1, 1)}) = w_H((g_{e_0} \oplus l_{(a, 1)})|_{(\mathcal{C}'_1, 1)}) = 2^{s-3} - 2^{\frac{s-3}{2}}.$$

Since, $g_e \oplus l_{(a, 1)}$ is not balanced and $\widehat{\mathcal{W}}_{g_e \oplus l_{(a, 1)}}(\bar{0}) = 2^{\frac{s-1}{2}}$ in $\mathcal{C}'_1|\bar{1}$.

Hence, adding the four cases,

$$|f^{-1}_{|\{(x_0, \bar{x}) \in \mathbb{F}_2^{s+1} | w_H(x_0, \bar{x}) \text{ odd}\}}(1)| = 2^{s-1}.$$

Therefore, $f_{|\{(x_0, \bar{x}) | w_H((x_0, \bar{x})) \text{ odd}\}}$ is balanced.

Resolving similarly, we obtained the same result in the other cases.

□

By analysing all the cases related to the demonstrations of Theorems 10 and 11, including those that remain unsolved, we derive Algorithms 1 and 2. In these algorithms, we define sets $\mathcal{A}$ and $\mathcal{B}$, where set $\mathcal{A}$ contains vectors of length o with even Hamming weights, and set $\mathcal{B}$ consists of vectors of length o with odd Hamming weights. Also, $l_a(x) := a_1x_1 \oplus \cdots \oplus a_rx_r$, $a = (a_1, \dots, a_r)$, $x = (x_1, \dots, x_r) \in \mathbb{F}_2^r$, for any positive integer r .

In the first algorithm, we derive bent functions for any even dimension in its domain that exceeds the dimension of the given bent function. These newly created bent functions are explicitly balanced when the domain is restricted to vectors with even Hamming weight. In the second algorithm, we examine the sum of a linear function and identify specific conditions for its balancedness.

Algorithm 1 Maiorana-McFarland secondary construction $r = 1$

Input: $s-1 \geq 2$ even, $g_{s-1} : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function, $x'^{s-1}, x''^{s-1} \in \mathbb{F}_2, x^{s-1} \in \mathbb{F}_2^{s-1}$

Output: g_o a bent function, $g_o|_{\mathcal{A}}$ balanced

```

1: Integer  $o$ ;
2:  $n := s - 1$ ;
3: while  $n < o$  do
4:   for  $x'^n, x''^n$  from 0 to 1 do
5:     if  $(w_H(x^n)$  is even,  $x'^n = 1, x''^n = 1$ ) or  $(w_H(x^n)$  is odd,  $x'^n = 1, x''^n = 0$ )
6:        $g_{n+2}(x'^n, x^n, x''^n) = 1 \oplus g_n(x^n)$ ;
7:     else
8:        $g_{n+2}(x'^n, x^n, x''^n) = g_n(x^n)$ ;
9:     end if
10:   end for
11:    $x^n = (x'^n, x^n, x''^n); \quad n := n + 2$ ;
12: end while

```

5 Conclusions

We have provided an alternative proof for the balancing of a bent function when its domain is restricted to vectors with even or odd Hamming weights. This result was crucial in demonstrating our direct proof for identifying bent functions on affine spaces. Consequently, we were able to derive a family of functions using the Maiorana-McFarland secondary construction, ensuring that their balancing is strict for vectors with even Hamming weights.

In this study, we analysed specific values within the domain of the Walsh-Hadamard transform of a bent function g in relation to the newly obtained bent functions. This analysis encompassed the functions on affine spaces, g_{e_1} and g_{e_2} , as well as the bent function derived from the Maiorana-McFarland secondary construction. Additionally, we examined the balancing properties, confirming that the bent functions obtained remain balanced within the set of

Algorithm 2 Maiorana-McFarland secondary construction $\oplus$ linear $r = 1$

Input: $s - 1 \geq 2$ even, $g_{s-1} : \mathbb{F}_2^{s-1} \rightarrow \mathbb{F}_2$ be a bent function, $l_{a^{s-1}}$ a linear function on $\mathbb{F}_2^{s-1}$, $x'^{s-1}, a'^{s-1}, x''^{s-1}, a''^{s-1} \in \mathbb{F}_2$, $x^{s-1}, a^{s-1} \in \mathbb{F}_2^{s-1}$

Output: $g_o \oplus l_{(a'^{o-2}, a^{o-2}, a''^{o-2})}$ a bent function. If $(a'^{o-2} = 0 \wedge a''^{o-2} = 0) \vee (a'^{o-2} = 1 \wedge a''^{o-2} = 1)$, $(g_o \oplus l_{(a'^{o-2}, a^{o-2}, a''^{o-2})})|_{\mathcal{A}}$ balanced. If $(a'^{o-2} = 0 \wedge a''^{o-2} = 1) \vee (a'^{o-2} = 1 \wedge a''^{o-2} = 0)$, $(g_o \oplus l_{(a'^{o-2}, a^{o-2}, a''^{o-2})})|_{\mathcal{B}}$ balanced

- 1: Integer o ;
- 2: $n := s - 1$;
- 3: **while** $n < o$ **do**
- 4: **if** $a'^n = 0$ **and** $a''^n = 0$ **then**
- 5: **for** x'^n, x''^n from 0 to 1 **do**
- 6: **if** $(w_H(x^n)$ is even, $x'^n = 1, x''^n = 1$) or $(w_H(x^n)$ is odd, $x'^n = 1, x''^n = 0$)
- 7: $(g_{n+2} \oplus l_{(0, a^n, 0)})(x'^n, x^n, x''^n) = 1 \oplus (g_n \oplus l_{a^n})(x^n)$;
- 8: **else**
- 9: $(g_{n+2} \oplus l_{(0, a^n, 0)})(x'^n, x^n, x''^n) = (g_n \oplus l_{a^n})(x^n)$;
- 10: **end if**
- 11: **end for**
- 12: $x^n = (x'^n, x^n, x''^n)$; $a^n = (a'^n, a^n, a''^n)$; $n := n + 2$;
- 13: **end if**
- 14: **if** $a'^n = 0$ **and** $a''^n = 1$ **then**
- 15: **for** x'^n, x''^n from 0 to 1 **do**
- 16: **if** $(w_H(x^n)$ is even, $x'^n = 0, x''^n = 1$) or $\{(w_H(x^n)$ is odd, $[(x'^n = 0, x''^n = 1)$
- 17: or $(x'^n = 1, x''^n = 0)$ or $(x'^n = 1, x''^n = 1)]\}$ **then**
- 18: $(g_{n+2} \oplus l_{(0, a^n, 1)})(x'^n, x^n, x''^n) = 1 \oplus (g_n \oplus l_{a^n})(x^n)$;
- 19: **else**
- 20: $(g_{n+2} \oplus l_{(0, a^n, 1)})(x'^n, x^n, x''^n) = (g_n \oplus l_{a^n})(x^n)$;
- 21: **end if**
- 22: **end for**
- 23: $x^n = (x'^n, x^n, x''^n)$; $a^n = (a'^n, a^n, a''^n)$; $n := n + 2$;
- 24: **end if**
- 25: **if** $a'^n = 1$ **and** $a''^n = 0$ **then**
- 26: **for** x'^n, x''^n from 0 to 1 **do**
- 27: **if** $(w_H(x^n)$ is even, $x'^n = 1, x''^n = 0$) or $(w_H(x^n)$ is odd, $x'^n = 1, x''^n = 1$)
- 28: $(g_{n+2} \oplus l_{(1, a^n, 0)})(x'^n, x^n, x''^n) = 1 \oplus (g_n \oplus l_{a^n})(x^n)$;
- 29: **else**
- 30: $(g_{n+2} \oplus l_{(1, a^n, 0)})(x'^n, x^n, x''^n) = (g_n \oplus l_{a^n})(x^n)$;
- 31: **end if**
- 32: **end for**
- 33: $x^n = (x'^n, x^n, x''^n)$; $a^n = (a'^n, a^n, a''^n)$; $n := n + 2$;
- 34: **end if**
- 35: **if** $a'^n = 1$ **and** $a''^n = 1$ **then**
- 36: **for** x'^n, x''^n from 0 to 1 **do**
- 37: **if** $\{w_H(x^n)$ is even, $[(x'^n = 0, x''^n = 1)$ or $(x'^n = 1, x''^n = 0)$ or $(x'^n = 1,$
- 38: $x''^n = 1)]\}$ or $(w_H(x^n)$ is odd, $x'^n = 0, x''^n = 1)$ **then**
- 39: $(g_{n+2} \oplus l_{(1, a^n, 1)})(x'^n, x^n, x''^n) = 1 \oplus (g_n \oplus l_{a^n})(x^n)$;
- 40: **else**
- 41: $(g_{n+2} \oplus l_{(1, a^n, 1)})(x'^n, x^n, x''^n) = (g_n \oplus l_{a^n})(x^n)$;
- 42: **end if**
- 43: **end for**
- 44: $x^n = (x'^n, x^n, x''^n)$; $a^n = (a'^n, a^n, a''^n)$; $n := n + 2$;
- 45: **end if**
- 46: **end while**

vectors possessing even Hamming weights. This analysis was also extended to include a linear function applied to the original bent function.

Based on our analysis, results, and the algorithms developed, this research may be significant for future investigations aimed at discovering Boolean functions that exhibit total balancedness and high non-linearity, starting from bent functions, as previously mentioned in [13]. It is important to note that we already possess half of the balanced domain concerning the images. For future research, our objective will be to obtain the other half while minimizing the reduction in the non-linearity of the bent function as much as possible.

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